

Instructor Key: Day 8 Mastery Quiz: Debugging and Test Design

Debugging and Test Design

Key for the Week 2 student quiz package. Keep instructor-facing.

1 Artifact status and source boundary

Field	Status
Student artifact	10-point Day 8 mastery quiz; quiz, not workbooklet.
Content anchor	Built from the locally verified Week 2 workbooklet content anchor for sensor-kit checkout and release routing.
Difficulty posture	Shorter and sharper than the workbooklet. It asks for trace, route, debug, test, and transfer evidence.
Student-surface rule	Student PDF hides internal file versioning and release/validation language; visible student surface carries Lance copyright.
Canvas status	Not Canvas-released. Student PDF still needs Lance upload/staging plus Ally and Panorama review if selected for use.

2 Reference code

```
1 charge_percent = 78
2 calibration_label = "current"
3 damage_note = ""
4
5 if damage_note != "":
6     route = "repair bench"
7 elif charge_percent > 80:
8     route = "charge station"
9 elif calibration_label != "current":
10    route = "calibration bench"
11 else:
12    route = "release kit"
13
14 print(route)
```

3 Point map, secure answers, and support moves

Pt.	Item target	Secure answer	Support move
1	1. expected route	"charge station" because 78 is below the minimum charge threshold.	Revisit source condition, first-true order, expected/actual evidence, or boundary case as appropriate.
1	2. actual route	"release kit" because damage is false, charge_percent > 80 is false, calibration is current, so else runs.	Revisit source condition, first-true order, expected/actual evidence, or boundary case as appropriate.
1	3. bug condition	charge_percent > 80 is the wrong direction for the charge-station hold.	Revisit source condition, first-true order, expected/actual evidence, or boundary case as appropriate.
1	4. minimal fix	Use charge_percent < 80 for the charge station route.	Revisit source condition, first-true order, expected/actual evidence, or boundary case as appropriate.
1	5. why not else	The else should mean no hold conditions are true; changing it would hide the real faulty charge condition.	Revisit source condition, first-true order, expected/actual evidence, or boundary case as appropriate.
1	6. boundary test	Test 79, 80, and/or 81. A secure answer states expected route for the chosen boundary.	Revisit source condition, first-true order, expected/actual evidence, or boundary case as appropriate.
1	7. regression clean release	Example: damage blank, charge_percent 82, calibration current, initials present should release or continue as clean according to the route map.	Revisit source condition, first-true order, expected/actual evidence, or boundary case as appropriate.
1	8. eighty case	At 80 the kit should not go to charge station if the intended hold is below 80.	Revisit source condition, first-true order, expected/actual evidence, or boundary case as appropriate.
1	9. expected-actual sentence	Expected charge station for 78; actual release kit; likely cause is the reversed > comparison.	Revisit source condition, first-true order, expected/actual evidence, or boundary case as appropriate.
1	10. support route	Build an expected/actual/cause/fix/test row before editing code.	Revisit source condition, first-true order, expected/actual evidence, or boundary case as appropriate.

4 Mastery interpretation

Band	Point guide	Evidence pattern	Next support action
Secure	8–10 / 10	Student names the route evidence, controlling condition, and transfer/debug implication.	Move to the next daily packet or Exam 1/Week 3 bridge as appropriate.
Developing	5–7 / 10	Student can get final route but may not preserve condition order or expected/actual evidence.	Use a route ledger: condition, value, branch, action, reason.
Needs support	0–4 / 10	Student guesses route from scenario language or final output without tracing.	Return to first condition, true/false result, branch selected, and support note.

5 Release/accessibility notes

Local verification should check tagged PDF output, extractable text, page count, render sheets, answer-key separation, no private/protected markers, and the student-facing no-internal-surface rule. If selected for Canvas use, Lance uploads/stages the student PDF and retrieves both Ally and Panorama scores.